

RANKING TRANSFERS, THRESHOLDS DON'T: PRIOR-CORRECTED TEST-TIME ADAPTATION FOR AUDIO DEEPPFAKE DETECTION

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Abstract—Audio deepfake detectors generalise poorly across corpora. We show the failure is one of *operating point* — ranking largely transfers while the decision threshold does not — and that confident-quantile test-time adaptation (TTA) needs its tail budget to satisfy $q \leq \min(\pi, 1-\pi)$: a counting bound caps pseudo-label purity at $\min(1, \pi_{\text{real}}/q)$ for any detector. That holds where TTA is measured and almost nowhere else. On ASVspoof2021-DF (97.2% spoof) the published $q=0.3$ therefore cannot exceed 9.3% purity, and it drives three publicly released checkpoints from 98.8–99.1% accuracy to 62–65%; three of four modern TTA baselines fail likewise, and resampling In-the-Wild’s prevalence reproduces the collapse on demand. A label-free prior estimate restores all three (9/9 paired cells, $p=0.004$), but the prior is not identifiable from a model’s own outputs, so its estimate degrades exactly as the operating point does ($r=0.997$) — whereas shrinking q needs no estimate.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, label shift, cross-corpus generalisation.

I. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [20].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high **ROC-AUC while its accuracy at a fixed threshold collapses**. It still *ranks* clips correctly; its operating point is mis-calibrated. Because ranking transfers, the most-confident predictions on unlabeled target audio are trustworthy enough to adapt on at test time.

That observation is the starting point of a family of methods, not a method. The step every member of the family takes—label the extreme quantiles of the target scores and self-train on them—carries an assumption that is invisible on the benchmarks the field uses and fatal on the pools it deploys to: that the target pool is roughly class-balanced. Take the top and bottom 30% of a 97%-spoof pool and the “confidently bona fide” bucket is mechanically made of low-scoring *spoof*. Self-training then teaches the detector that a great deal of spoof is genuine. This is not hypothetical: it is what happens to all three released publicly released checkpoints we test on the

official ASVspoof2021-DF protocol (Table I), and to three of four modern TTA baselines.

Contributions. (i) The ranking–calibration asymmetry as the mechanism of cross-corpus failure, over 369 third-party cells and tested directly: a proxy $\mathcal{A}$ -distance *rises* after adaptation, so the effect is operating-point repair, not learned invariance. (ii) An exact *necessary* prevalence condition for the confident-quantile rule, $q \leq \min(\pi, 1-\pi)$, from a counting bound on pseudo-label purity; it accounts for every result here and is confirmed by intervening on prevalence directly. (iii) A label-free correction that removes the damage on every DF checkpoint, with an identifiability argument bounding when it can be trusted — and an oracle-prior arm showing that what fails is the estimate, not the correction. (iv) Two label-free monitors, one of which recovers 13.9 EER points from a Tent collapse out-of-sample, together with the precondition they themselves require.

Related work. SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]. Train-time domain generalisation aligns source domains [5] or runs gradient reversal over source and target jointly [13]; both act before deployment. In TTA, entropy minimisation (Tent) [7] is the reference recipe, and its instability prompted reliable-sample filtering (ETA/EATA) [8], sharpness-aware entropy with recovery (SAR) [9], and information maximisation with self-labelling (SHOT) [10] — we compare against all four. Closest to us is T²A [17], which likewise targets TTA for *deepfake detection* and likewise diagnoses entropy minimisation’s confirmation bias, replacing it with uncertainty-aware negative learning; it is evaluated on image/video deepfakes and does not consider the target class prior, which is the axis this paper isolates. Our findings are complementary: T²A repairs the objective, we show that the *sampling* of pseudo-labels carries a prevalence precondition that no objective can remove. Batch-statistics recalibration [14] is a no-op for LayerNorm. The prior estimator is black-box shift estimation [11]; the proxy $\mathcal{A}$ -distance is from [15].

II. METHOD

Let x be a waveform with label $y \in \{0, 1\}$ (fake= 1) and let f_θ produce $s_\theta(x) = P_\theta(y=1 | x)$. We are given a source

model, its labelled training corpus $\mathcal{S}$, and an *unlabelled* target pool $\mathcal{T}$. No target label is read at any point.

What breaks under corpus shift. Ranking quality $\text{AUC} = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free; accuracy at a fixed τ is not. Empirically AUC degrades far less than accuracy at $\tau=0.5$ under source→target shift: across 369 third-party evaluation cells the accuracy a model’s ranking supports but its shipped threshold does not deliver averages 7.29 points. The deployable object is therefore the score *order*, and the broken object is the operating point. Everything below follows from taking that literally.

(a) The target prior, label-free. An operating point is a statement about a prior, so we estimate the prior rather than assume it. Black-box shift estimation [11] gives $\hat{p} = M^{-\top} q$ with $M[y, \hat{y}] = P_{\mathcal{S}}(\hat{y} | y)$ the source confusion matrix and q the target prediction histogram. Estimators split into those reading the *thresholded prediction rate* (BBSE, EM [12]) and those reading the score *histogram* (a two-component mixture); the first inherits whatever calibration error the target induces, and Sec. IV show neither is reliable where it is needed.

(b) The tail budget, and when it is valid. The standard rule takes the bottom q and top q of the target scores as pseudo-real and pseudo-fake. Its validity has a characterisation that needs no model at all. A bucket of $q_c N$ clips can contain at most the pool’s $\pi_c N$ class- c clips, so *for any detector whatsoever*

$$\text{purity}_c \leq \min(1, \pi_c / q_c). \quad (1)$$

This is counting, not an approximation; what is empirical is its *tightness*, which is where the ranking precondition enters — across all seven (checkpoint, corpus) cells here the bound is never violated and is attained to within 0.025, the slack tracking AUC (0.000 at AUC .985, 0.025 at .917). Two consequences follow, and they are the whole story.

The *symmetric* rule sets $q_c = q$, so perfect purity is *feasible* only while

$$q \leq \min(\pi, 1 - \pi). \quad (2)$$

This is *necessary, not sufficient*: violating it makes pure pseudo-labels impossible for any detector, while satisfying it only removes the prevalence obstruction and leaves purity to the ranking. Our claims rest on the necessary direction, which is where the published configurations fall. In-the-Wild satisfies this at $q=0.3$ (min=.372) and the recipe repairs checkpoints there; DF does not (min=.028), where no detector can exceed 9.3% pseudo-real purity — we measure 8.6% — and the recipe destroys them. The damage is one-sided: DF’s pseudo-*fake* tail is 100% pure at every q .

The *prior-split* rule, $q_c = 2q\hat{\pi}_c$, restores Eq. (1) for *any* $q \leq \frac{1}{2}$ given a correct prior, and is identical to the symmetric rule at $\hat{\pi}=0.5$. With an estimate it needs

$$\hat{\pi}_c / \pi_c \leq 1/(2q) \quad \text{for both classes}, \quad (3)$$

i.e. at $q=0.3$ the estimate may overstate either class by at most $1.67\times$. This is why it works on DF ($\hat{\pi}_{\text{real}}/\pi_{\text{real}}=1.03$) and fails on In-the-Wild s42, where underestimating π_{real} inflates the *fake* budget to 0.587 against $\pi_{\text{fake}}=0.372$ ($2.63\times$, bound

0.633) and simply moves the impurity to the other tail. Across the seven adapted arms we ran, the worst-tail bound is 1.000 for the five that came out intact (82–99% accuracy) and 0.093 and 0.633 for the two that collapsed (62.96, 47.99). Since q is ours to choose and π is not (Sec. IV), a third option is to shrink q until Eq. (2) holds for the most skewed prior anticipated — at a supervision cost the prior-split budget avoids ($q=0.02$ labels 800 of 20,000 clips against 12,000). A fixed confidence threshold is a fourth, prior-free but calibration-dependent. We report all four.

(c) Objective. With pseudo-labels $\tilde{y}$ from (b) on the confident set $\mathcal{C}$ and a stochastic channel perturbation $a(\cdot)$ (per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus Gaussian noise $\sigma=0.005$),

$$\min_{\{\gamma, \beta\}, \phi} \underbrace{\sum_{x \in \mathcal{C}} \text{CE}(f_\theta(x), \tilde{y})}_{\mathcal{L}_{\text{ST}}} + \lambda \underbrace{\|f_\theta(a(x)) - \text{sg}[f_\theta(x)]\|^2}_{\mathcal{L}_{\text{cons}}}, \quad (4)$$

optimising only the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned blocks and the head ϕ (16,706 parameters), $\lambda=0.3$. Restricting the trainable set and using only prior-corrected confident tails is what prevents the degenerate collapse entropy minimisation is prone to. The consistency term is inherited from the published recipe; an ablation finds it contributes nothing here (3.98 vs 4.02 EER), so the budget is doing the work.

III. EXPERIMENTAL SETUP

Protocol A. The official ASVspoof2021-DF eval partition, scored against the official CM keys. Of its 611,829 trials we hold 400,435; the absent shard is uniform across codec, label and phase (every stratum splits 75–25 present/absent, matching the overall ratio), so the EER we pool over those 400,435 is *not* the official full-set figure but an unbiased estimate of it. True spoof rate 97.22%. Published pooled EERs span 1.92 [2], 1.95 [18] and 2.85 [1] down to the challenge-era best of 15.64 [19]; our own source model reaches only 26.33, so we run the method on *released* checkpoints instead: DeepFense XLS-R 300M + AASIST trained on ASVspoof2019-LA [25], three independent training seeds. Adaptation uses an unlabelled 20,000-clip subsample; we report the full official pooled EER, an inductive number over trials *not* in that subsample, and two further seeds on a fixed 50,000 subset, giving nine paired cells.

Protocol B. Leave-one-corpus-out over ASVspoof2019-LA [21], D2 (LibriSpeech vs. eight open TTS systems), In-the-Wild [20] and the Arabic set [23]; each source pool merges the other three with a 38-language MLAAD subset [22], held-out target audio never appears in any source pool, and six MLAAD languages are held out of every pool. All methods run on the same twelve folds. **Protocol C** is all 31,779 In-the-Wild clips at their native 37.2% fake rate, skewed the *opposite* way to DF.

Comparison points. Source-only; the median-threshold rule; Tent, SHOT, ETA/EATA, SAR; AdaBN [14]; DANN [13] (which sees the held-out target’s unlabeled audio every step); an ASDG-style baseline [5]; and a target-supervised oracle.

Every TTA arm adapts the same 16,706 parameters with the same lr, batch and pass count; none was tuned separately, and none could be, since per-target tuning needs the labels the setting assumes absent. Two published constants are degenerate at $C=2$ and were re-derived: EATA’s redundancy filter (unreachable until the running mean hits a simplex vertex, after which only minority-predicted samples pass) and SAR’s recovery trigger ($e_0=0.2$ sits inside the $C=2$ reliable band and freezes the model; we keep its relative position, $e_0=(0.2/\ln 1000)\ln C$).

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning for our own source models (Adam, lr 2×10^{-4} , batch 32, 8 epochs). Adaptation: lr 1×10^{-4} , $E=4$ passes, batch 32 (third-party crops matched to each checkpoint’s own training length). Code and manifests: github.com/RohamIzadidoost/Kaggle_deepFakeAudio (branch `icassp-breadth`), with per-arm score dumps that reproduce every number here without a GPU.

IV. RESULTS

A. The class prior is the hidden variable

Table I is the main result, and Eqs. (1)–(2) anticipate it. Applied unmodified to a released checkpoint already better than the published Wav2Vec2-AASIST baseline here, the symmetric-quantile configuration *damages* it: accuracy $98.87 \rightarrow 62.96\%$, EER $4.51 \rightarrow 5.84\%$, manufacturing a 31-point deficit where there was none.

The published baselines fail likewise, each through its own version of the assumption. SHOT minimises $\mathbb{E}[H(p)] - H(\mathbb{E}[p])$, whose diversity term is maximised when the mean prediction is *uniform* — at $C=2$ a hard prior of $\frac{1}{2}$; the regulariser introduced to *cure* Tent-style collapse encodes the assumption more literally than Tent does. ETA is more instructive: its reliability filter, meant to discard the high-entropy samples that destabilise entropy minimisation, admitted 99.8% of them, so ETA reduced to Tent and landed worse. An entropy-keyed filter separates confident from uncertain, not *confident-and-right* from *confident-and-wrong*, and miscalibration manufactures the latter. SAR is the exception that clarifies the rule: the only prior method that does not degrade the checkpoint, achieved by declining to move it (308 recovery resets in 2,500 steps). On the other two checkpoints ETA reaches 31.0 and 37.5% EER (source 3.8, 4.6) and Tent 17.6 and 11.2. So the claim is not that every published method fails here, but that *three of five prior configurations degrade a strong released detector at the standard benchmark’s own prevalence, the fourth avoids damage only by declining to adapt, and none improves it*.

The prior-split budget reverses this: BBSE returns $\hat{\pi}=0.9715$ against a true 0.9708, comfortably inside Eq. (3), and EER improves to 4.02%. The change between the two arms is one scalar. Over three independently trained checkpoints (EER % / accuracy %):

Over the nine paired cells (seed-to-seed EER std ≤ 0.14), prior-correction beats the symmetric budget on 9/9 (+1.23 EER, Wilcoxon $p=0.004$) and source-only on 8/9 (+0.48,

$p=0.027$), while the symmetric budget is worse than source-only on 6/9. Accuracy separates them with no overlap at all: 98.8–99.1% (source), 61.6–65.2 (symmetric), 98.7–99.0 (prior-corrected). What does *not* replicate is the symmetric budget’s effect on *ranking* (EER worse on two checkpoints, better on one), as Eq. (1) implies: a mis-specified budget corrupts the pseudo-labels and hence the operating point, while the ranking is inherited.

B. The failure, and where the recipe already works

At In-the-Wild’s native prior (0.372) Eq. (2) *holds*, and the same recipe repairs rather than destroys. DeepFense s42 — the checkpoint that scores 4.56% on DF — arrives here ranking at AUC 0.917 while correct on only 39.1% of clips, fewer than a majority-class predictor. Self-training repairs exactly that: over four released checkpoints it drives the deficit from 16.7, 22.6 and 44.4 to 0.1, 2.9 and 5.0 (and costs a little on the fourth, which arrives well-calibrated at 2.1 and has nothing to repair). On s42 it also raises AUC .917 $\rightarrow$.947 — which no threshold can do — with the disjoint half slightly better (12.27/.949), so the ranking gain is not memorisation.

C. Testing the bound by intervention

Sweeping q on DF s2 with everything else fixed gives accuracy 98.96, 95.68, 88.15, 62.96 at $q=0.02, 0.05, 0.10, 0.30$ against bounds 1.000, 0.556, 0.278, 0.093 — monotone in the bound (Spearman +1.000). At $q=0.02$, inside Eq. (2) for this pool, the *unmodified symmetric rule* is safe and improves on source (4.51 $\rightarrow$ 4.19) — no prior estimate anywhere. It remains weaker than the prior-split budget (4.02) for the reason Eq. (1) implies: shrinking q fixes a minority-tail problem by discarding the majority tail too (800 of 20,000 clips labelled, against 12,000).

Inducing the failure where the recipe works. DF is one benchmark that happens to be 97.2% spoof, so we intervene on prevalence directly: resample In-the-Wild — where $q=0.3$ satisfies Eq. (2) and repairs checkpoints — to a target $P(\text{fake})$, holding audio and weights fixed. Accuracy at $q=0.02/0.10/0.30$ is 97.8/95.9/73.3 at $P(\text{fake})=0.90$, 98.6/92.6/68.5 at 0.95 and 98.6/89.4/65.4 at 0.97, against purity bounds of 1.00/1.00/0.33, 1.00/0.50/0.17 and 1.00/0.30/0.10.

Accuracy tracks the bound (Spearman +0.966): $q=0.3$ collapses monotonically, $q=0.10$ degrades from 0.95 exactly where its bound does, and $q=0.02$ is unharmed. The failure belongs to prevalence, not to DF. Necessity is what is tested: we make the condition fail, and the method fails with it.

Is the correction wrong, or the estimate? On the hardest cell (In-the-Wild s42, $\delta=44.4$), substituting the *true* prior gives 11.59 EER at 91.3% accuracy, against 12.58/82.4 for the symmetric budget and 14.51/48.0 and 14.95/55.0 for the BBSE and mixture estimates.

With the true prior the prior-split budget is the best method in this paper on its hardest cell; with either estimate it is worse than using no prior. The mechanism is sound and the estimation is not — exactly what Sec. IV proves must happen.

TABLE I: ASVspoof2021-DF eval, the 400,435 of 611,829 official trials we hold, 97.22% spoof, released XLS-R+AASIST checkpoint seed 2 [25]; every arm was also run on seeds 42 and 240 (text). All adaptation is label-free. Deficit $\delta = (100 - \text{EER}) - \text{acc}@0.5$ is the accuracy the ranking supports but the threshold does not deliver. ρ and the prediction-rate gap are the label-free monitors of Sec. II(d) — computable at deployment, unlike every other column.

	EER %	AUC	acc@0.5	deficit	ρ	gap
Source (as released)	4.51	.9923	98.87	-3.38	1.000	.000
+ ETA [8]	7.44 [‡]	.9288	98.93	-6.37	0.312	.007
+ Tent [7]	4.06 [‡]	.9765	98.52	-2.58	0.438	.013
+ SHOT [10]	6.38	.9725	77.64	+15.99	0.767	.225
+ TTA, fixed confidence 0.95	4.30	.9924	99.02	-3.32	0.827	.005
+ TTA, symmetric $q=0.3$	5.84	.9846	62.96	+31.20	0.816	.372
+ TTA, prior-corrected	4.02	.9931	98.95	-2.97	0.867	.007
+ SAR [9]	4.47 [§]	.9923	98.88	-3.35	0.970	.000

[‡] EER over a pool that is more than 93% a single tied score (Tent saturates 98.07% of scores to exactly 1.0, ETA 97.71%); read AUC. [§] SAR neither damages nor adapts: 308 recovery resets in 2,500 steps, AUC unchanged to four decimals.

Not a re-thresholding result. Both EER and AUC move, and neither takes a threshold as input: the same prior estimate used as a threshold leaves EER at 4.51 by construction, and as a budget gives 4.02.

D. Cross-corpus, balanced pools

Where Eq. (2) holds the prior correction is inert by construction, so this arm measures the rest of the method. Over twelve identical folds we beat source-only (18.39 $\rightarrow$ 17.44 pooled EER, $p=0.027$), Tent (32.25), ETA (29.54), EATA (31.34) and SAR (18.39), all $p \leq 0.027$ — but **not SHOT** (17.76 vs. 17.44, 7/12, $p=0.47$), which beats us on ASVspoof2019. That is the point: SHOT’s diversity term is a hard uniform prior, free when Eq. (2) holds and destructive when it does not (6.38 EER, 77.6% accuracy on DF). The separation between the two methods lives entirely in the regime the field does not test.

E. Label-free guards, and their own precondition

Eq. (2) says when pure pseudo-labels are feasible, but checking it needs π , which Sec. IV shows is not identifiable. A deployer is left with a weaker question that *can* be asked without labels — not *will* this adaptation help but *did* it — and two statistics answer it: rank agreement with the frozen source, $\rho = \text{Spearman}(s_\theta, s_{\theta_0})$, and the drift of the predicted positive rate from $\hat{\pi}$ (Table I). On DF they work complementarily — ρ catches ranking collapse (ETA 0.31, Tent 0.44), drift catches operating-point collapse (0.37, 0.39, 0.23) and is blind to the former, and requiring both separates the five harmful arms from the three harmless ones for any $\tau \in [0.70, 0.80]$ — and as a leave-one-target-out *stopping* rule ρ recovers 13.9 EER points from Tent’s collapse (32.25 $\rightarrow$ 18.33, oracle 17.24) while never firing on our own method.

But the drift monitor has a precondition of its own, and it is the same δ the rest of the paper is about. Repairing a mis-calibrated operating point *requires* a large move in the predicted positive rate — that is what the repair is — so drift separates repair from damage only when the source already sits near the right rate. On In-the-Wild it inverts: the two largest drifts (0.20, 0.50) are the two largest *improvements* (+17, +43 accuracy points, at $\delta=16.7$ and 44.4), while the one harmful arm drifts least (0.04, $\delta=2.1$). A guard tuned on DF raises

twelve false alarms there, including aborting the best arm in this paper. The guards are sound where the source arrives well-calibrated ($\delta \approx 0$, as on DF) and unusable where it does not — the same regime, and the same reason, as the prior estimate itself.

F. The target prior is not identifiable

How far can a label-free prior estimate be trusted? Not far, for structural reasons rather than any defect of BBSE: such a method observes only $\Pr[\hat{y}=1] = \pi \text{TPR}_\tau + (1-\pi)\text{FPR}_\tau$ — one equation in three unknowns. BBSE closes it by assuming the source error rates carry over, which is exactly what covariate shift breaks, so an extreme prior and an extreme calibration failure are *indistinguishable* from a model’s outputs: DF/s2 predicts 97.4% fake at a true prior of 0.972, ITW/s42 predicts 98.1% at a true 0.372.

The consequence is measurable: the estimate degrades exactly as the operating point does, $|\hat{\pi} - \pi|$ against the source deficit giving $r=0.997$ over ten cells. Estimating the prior and detecting that the estimate is wrong are one problem, and the monitors inherit it: at our worst case ($\hat{\pi}=0.979$ against 0.372) the prior-split budget is harmed (14.51 EER against 12.58) but still beats the unadapted source (16.47).

This is why Eq. (2) is worth more than the estimator: it needs only a *lower bound* on $\min(\pi, 1-\pi)$, not the prior. Where the prior cannot be estimated, drop the prior-split budget and the prior-anchored monitor and shrink q ($q=0.02$ was safe on every cell, priors 0.372 to 0.972).

G. Breadth, and the mechanism

The third-party arm is 369 cells: nine released checkpoints (five architectures) $\times$ ten corpora, restricted to the 76 pairs whose checkpoint was not trained on that corpus, at one, three or ten seeds and counted once per (checkpoint, corpus, seed). The effect is *re-calibration*: δ falls 7.29 $\rightarrow$ 1.96 (248/369, $p=2.2 \times 10^{-17}$); the mechanism is not learned invariance — a proxy $\mathcal{A}$ -distance and a domain-origin probe both *rise* after adaptation (Arabic 1.545 $\rightarrow$ 1.606) — and beyond any threshold the gradient step adds separability where the ranking is strong (ASVspoof2019 within-attack-family AUC rises on all six families). **Limitations:** Eq. (2) is necessary not sufficient, the gain varies widely, and a low-AUC start we cannot rescue.

COMPLIANCE WITH ETHICAL STANDARDS

This work used only publicly released datasets and model checkpoints under their respective licences; no new human-subject data were collected and no human or animal experiments were performed. The authors declare no conflict of interest.

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